Shower Shape Distributions: Configuration Variations

April 6, 2026

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1 showershape_1p5rad

Changes from nominal

```diff
--- config_showershape.yaml
+++ config_showershape_1p5rad.yaml
@@ -20,5 +20,5 @@
   response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
   data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_showershape"
+  var_type: "bdt_showershape_1p5rad"
   final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"

@@ -239,10 +239,10 @@

   # Run range for data processing (-1 = no bound)
-  run_min: -1
-  run_max: -1
+  run_min: 51274
+  run_max: 54000
   # Optional run list file (one run number per line). If set, only listed runs are processed.
   run_list_file: ""
   # Total luminosity (pb^-1) for the selected run range
-  lumi: 49.562
+  lumi: 16.8588

   unfold:
```

1.1 Nominal

1.1.1 no nbkg cut

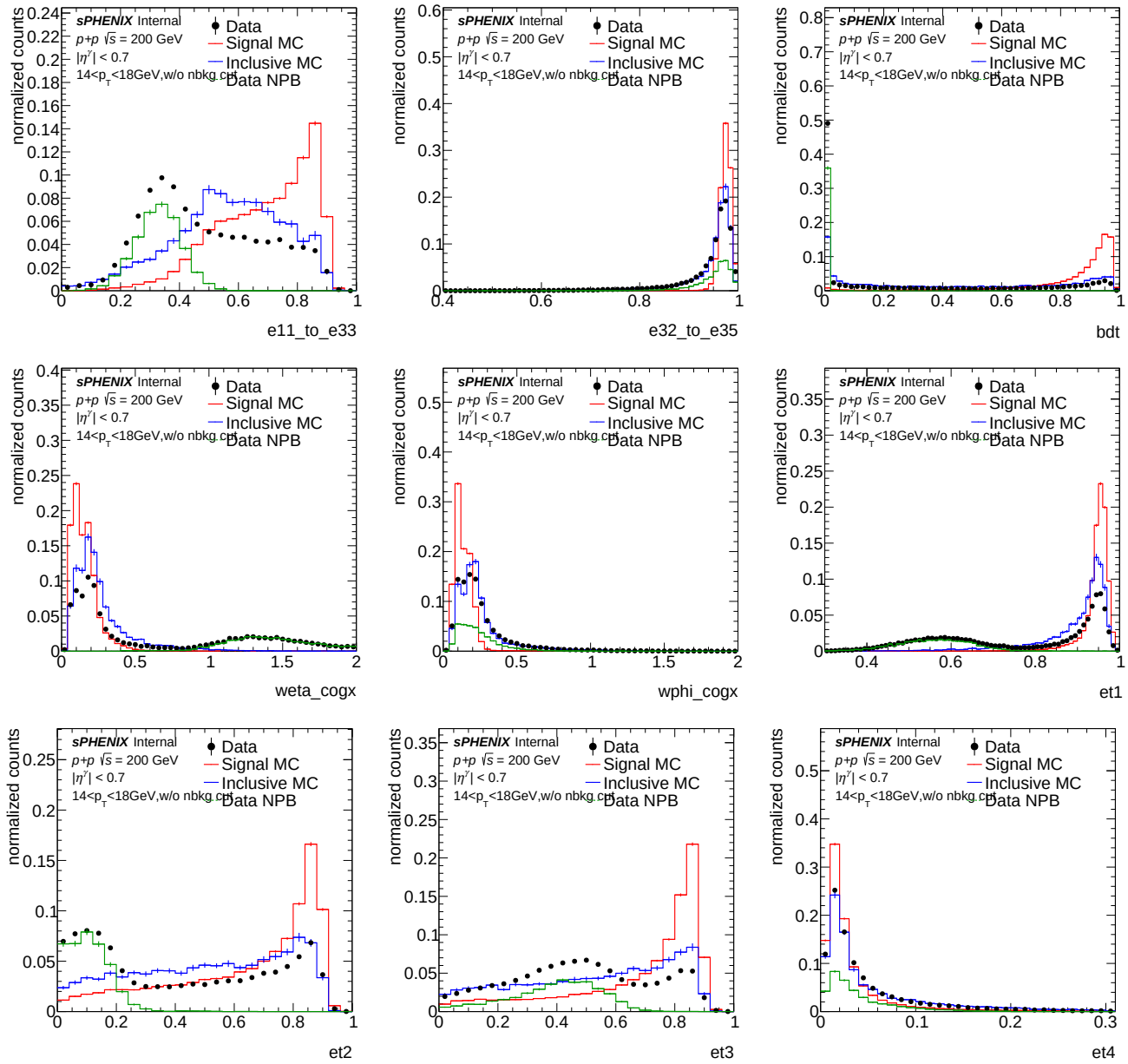

Figure 1: Shower shape distributions (Nominal, Data vs MC), $14 < E_T < 18$ GeV, no nbkg cut.

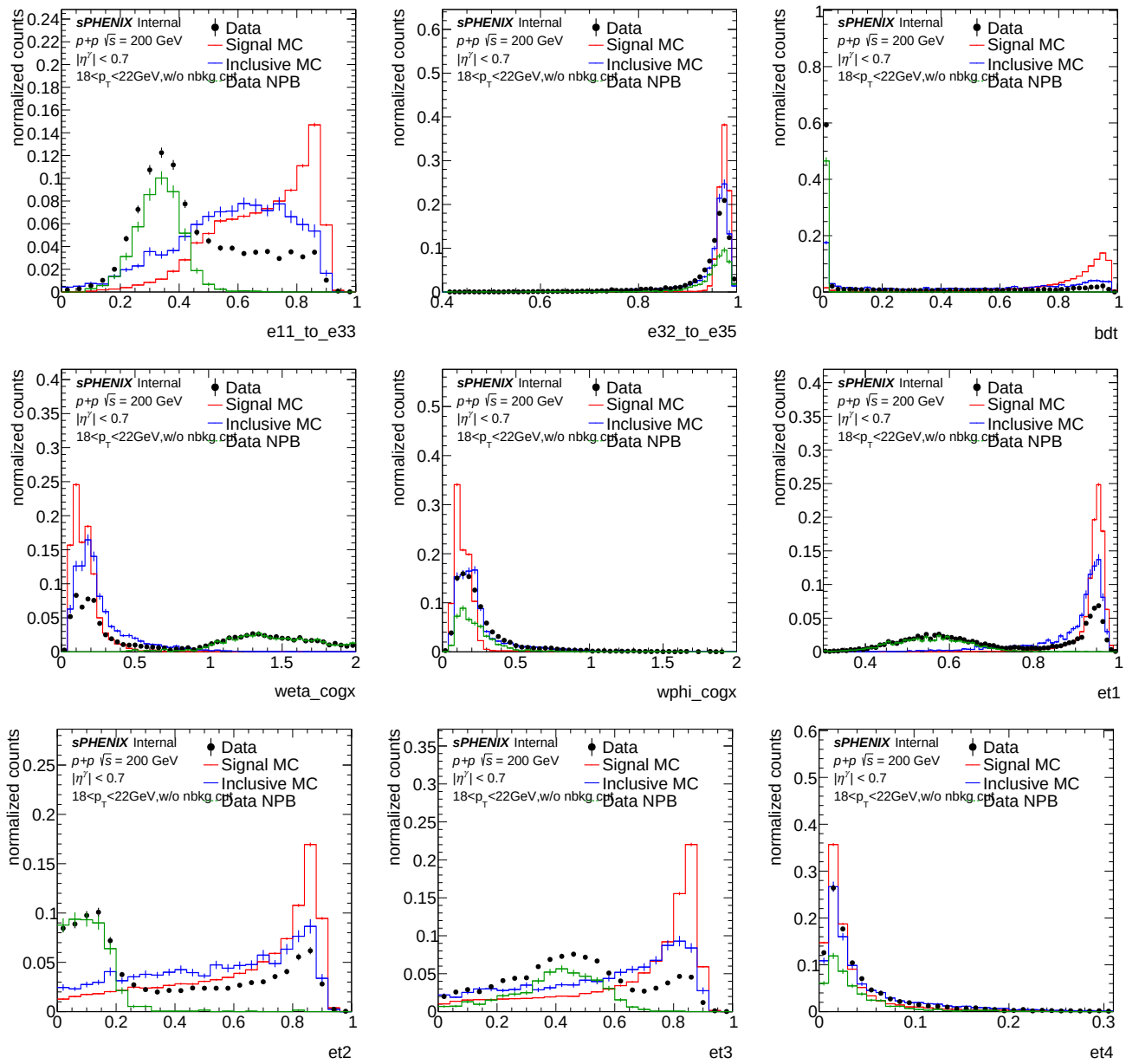

Figure 2: Shower shape distributions (Nominal, Data vs MC), $18 < E_T < 22$ GeV, no nbkg cut.

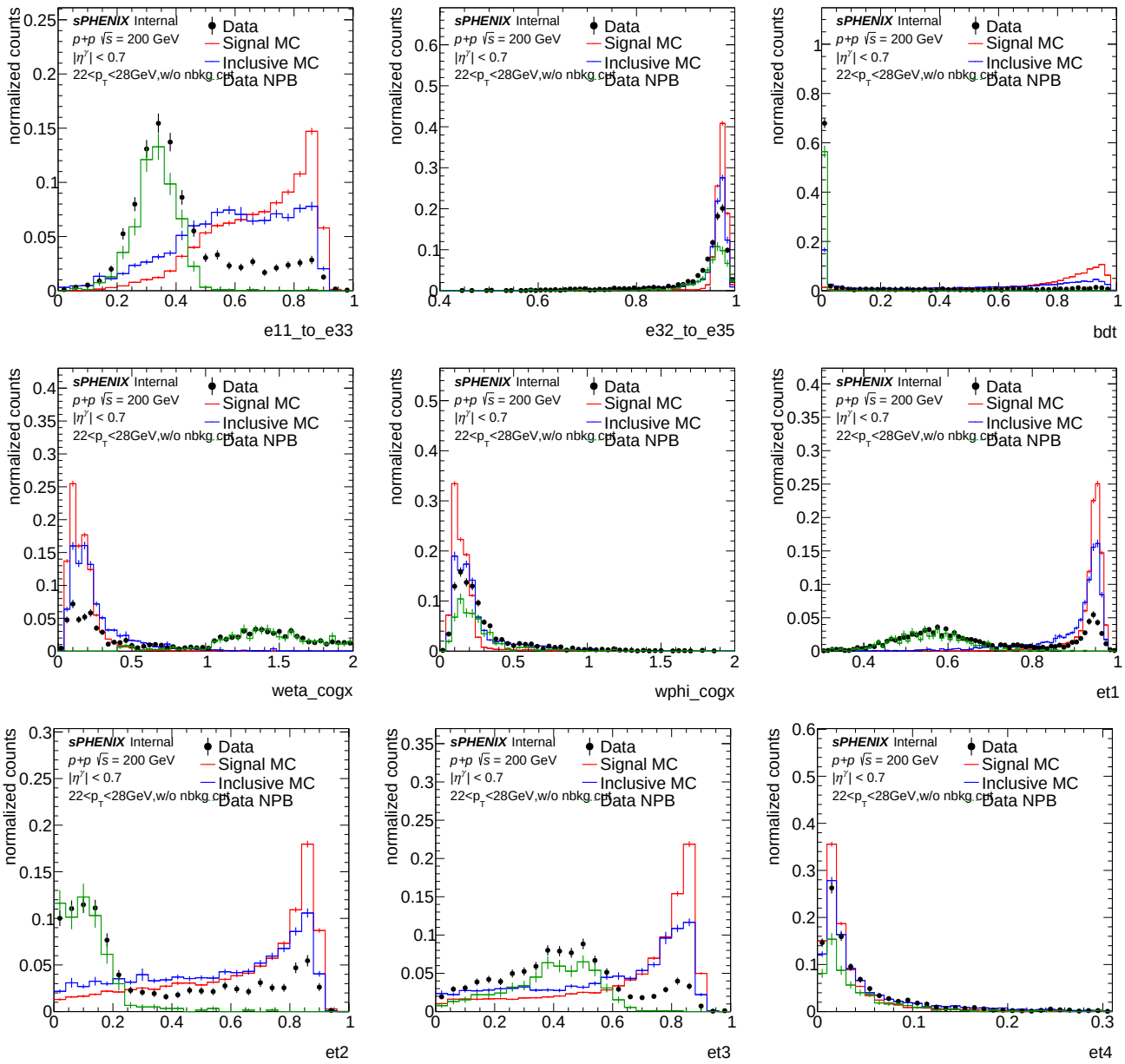

Figure 3: Shower shape distributions (Nominal, Data vs MC), $22 < E_T < 28$ GeV, no nbkg cut.

1.1.2 nbkg cut

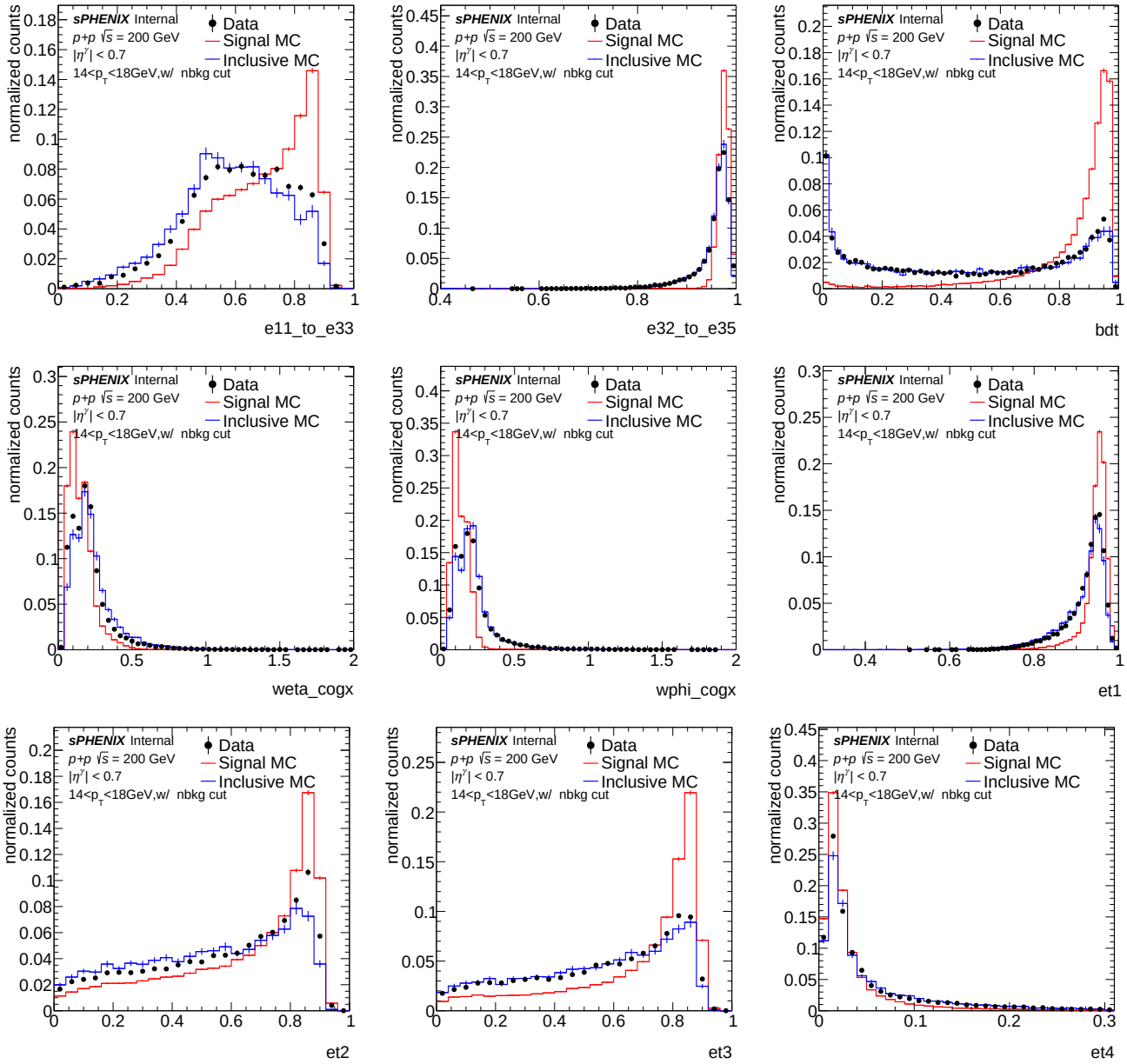

Figure 4: Shower shape distributions (Nominal, Data vs MC), $14 < E_T < 18$ GeV, nbkg cut.

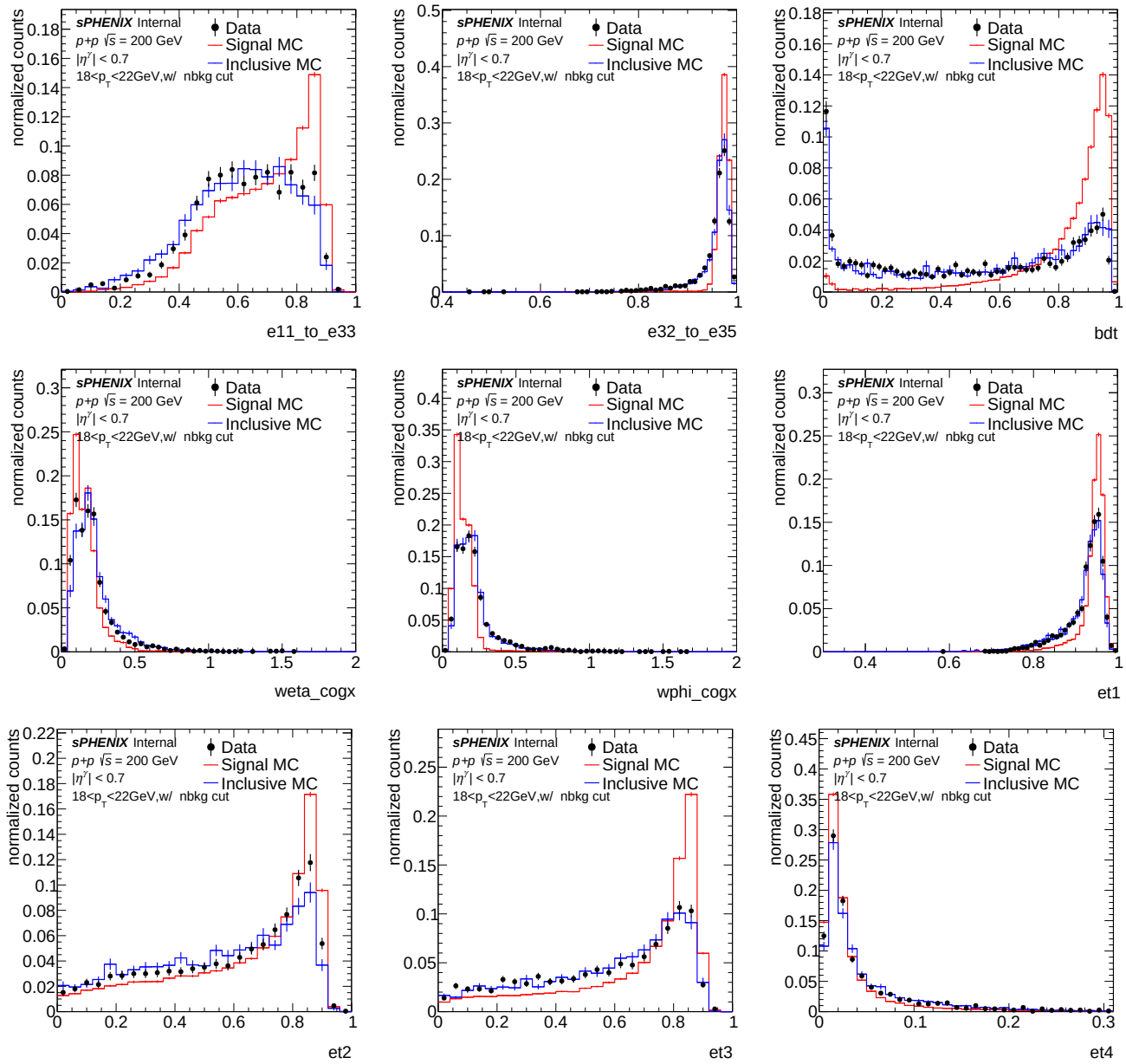

Figure 5: Shower shape distributions (Nominal, Data vs MC), $18 < E_T < 22$ GeV, nbkg cut.

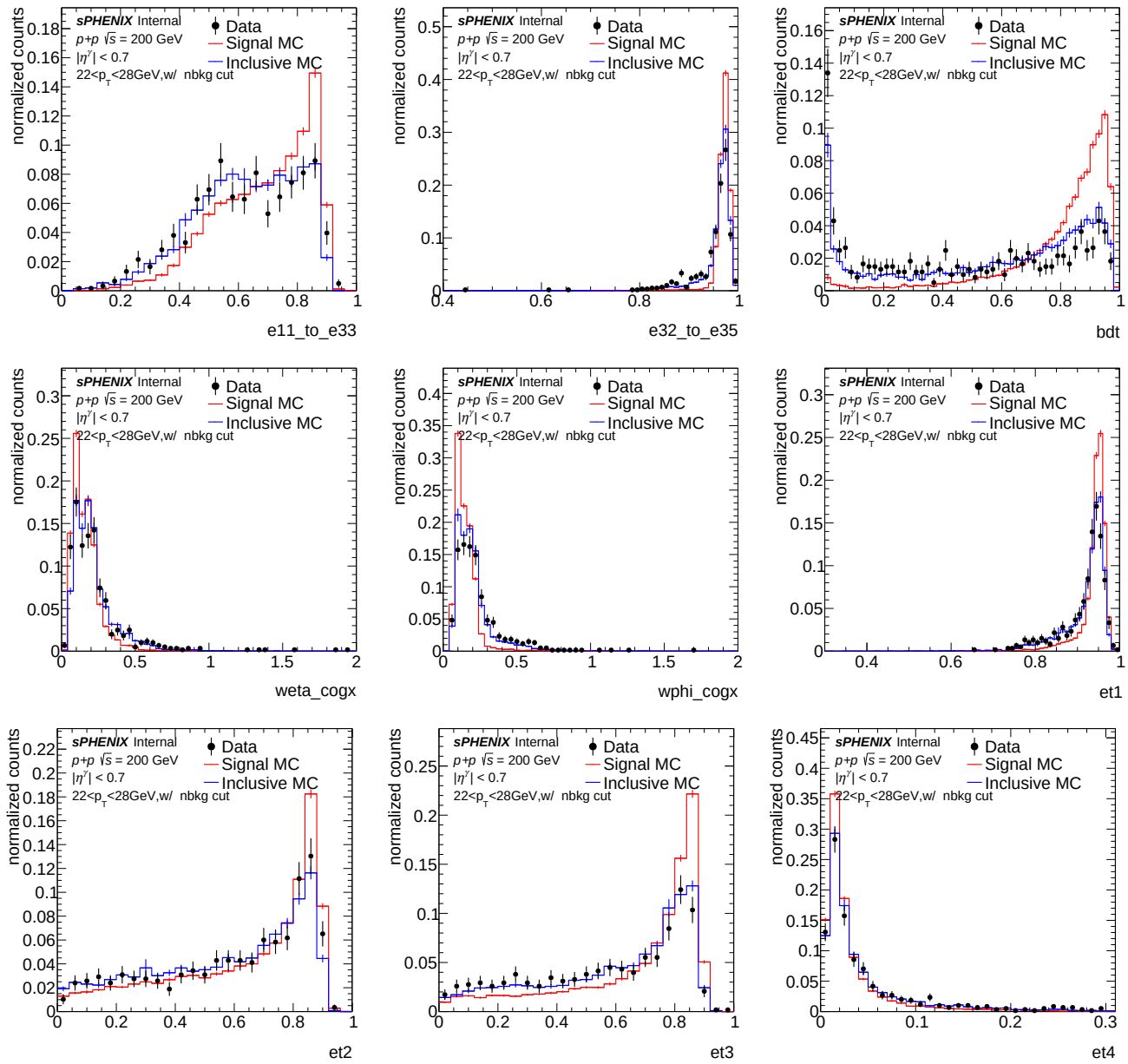

Figure 6: Shower shape distributions (Nominal, Data vs MC), $22 < E_T < 28$ GeV, nbkg cut.

1.1.3 tight photon ID

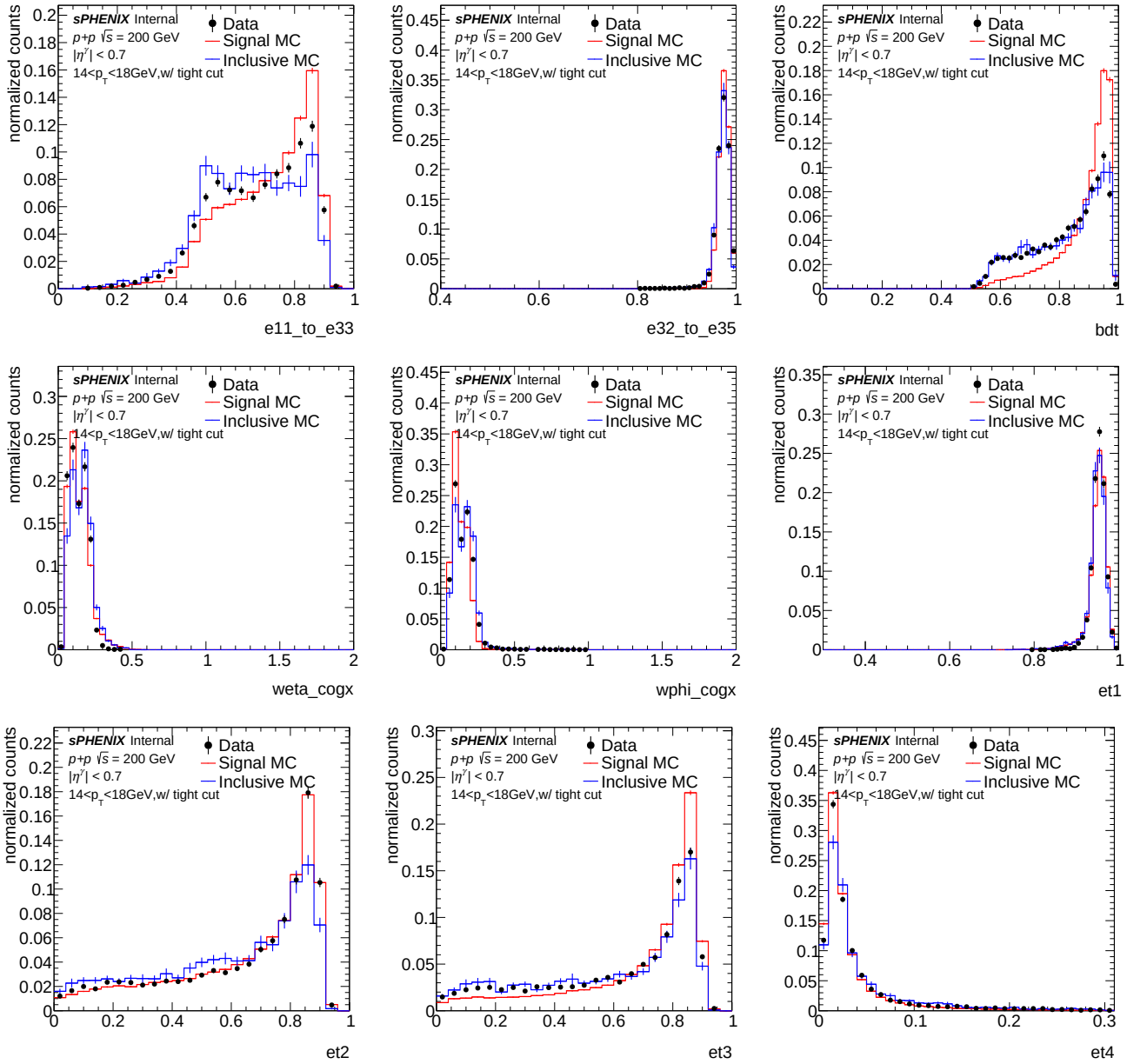

Figure 7: Shower shape distributions (Nominal, Data vs MC), $14 < E_T < 18$ GeV, tight photon ID.

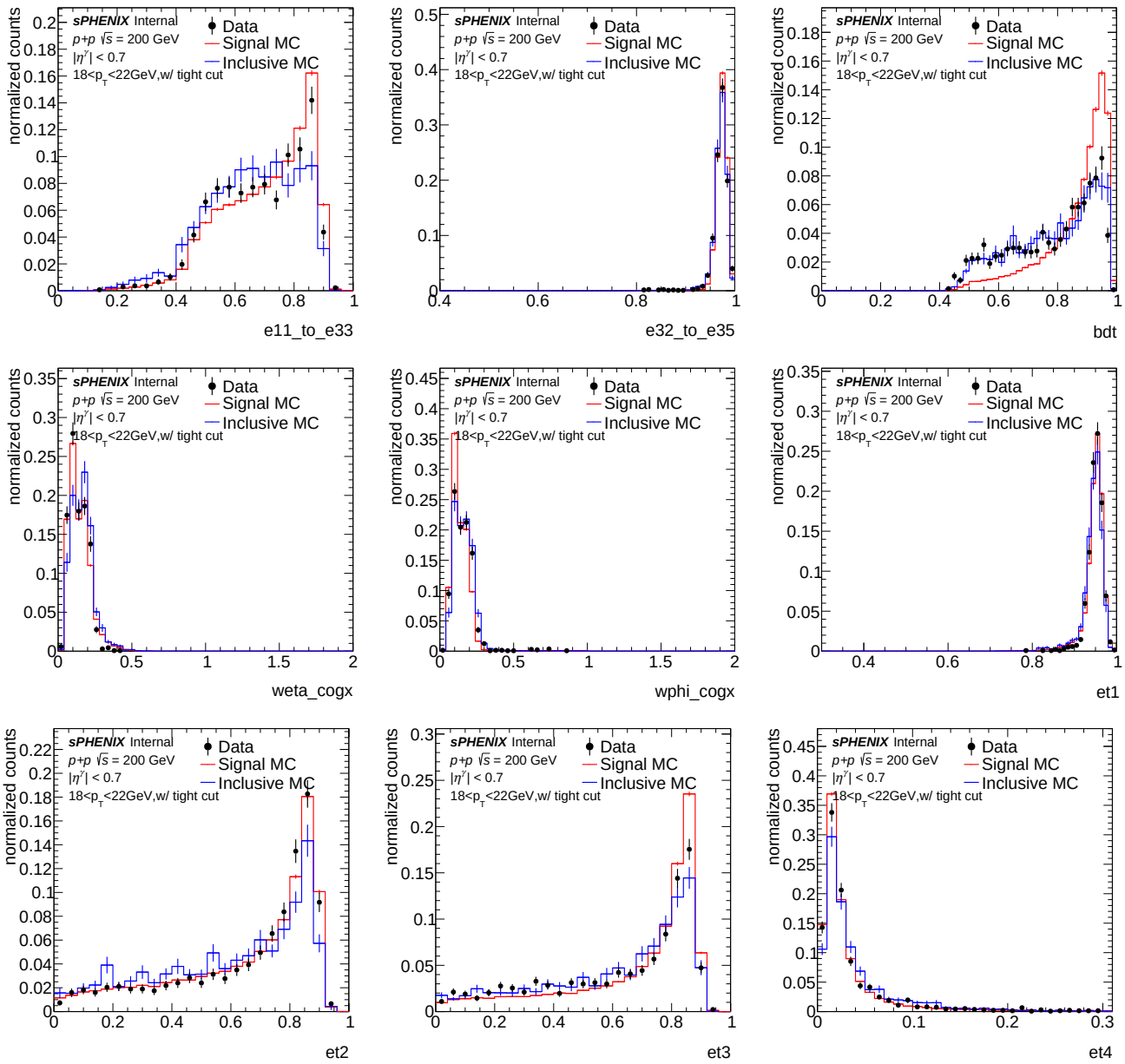

Figure 8: Shower shape distributions (Nominal, Data vs MC), $18 < E_T < 22$ GeV, tight photon ID.

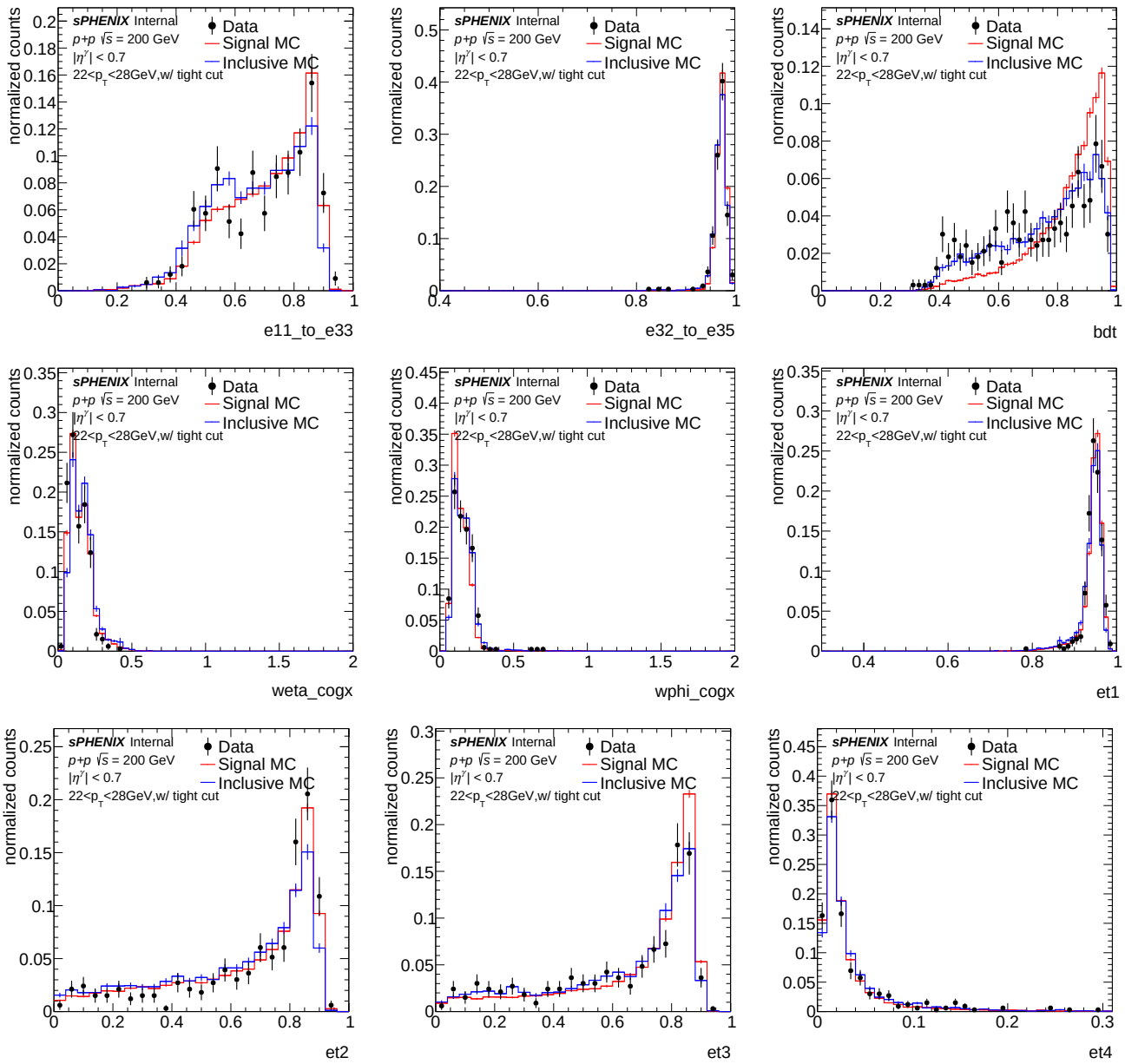

Figure 9: Shower shape distributions (Nominal, Data vs MC), $22 < E_T < 28$ GeV, tight photon ID.

1.1.4 non-tight (sideband)

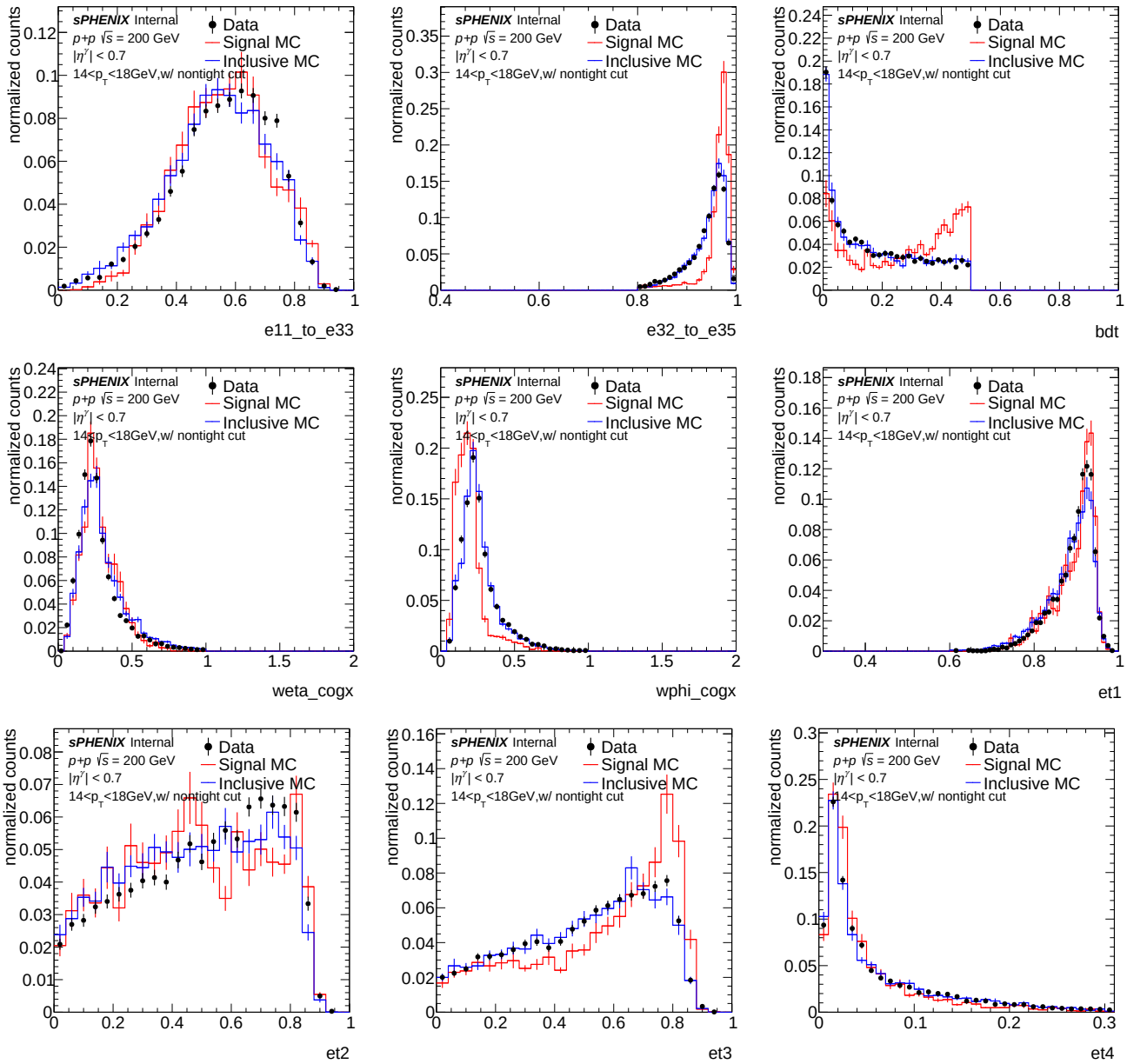

Figure 10: Shower shape distributions (Nominal, Data vs MC), $14 < E_T < 18$ GeV, non-tight (sideband).

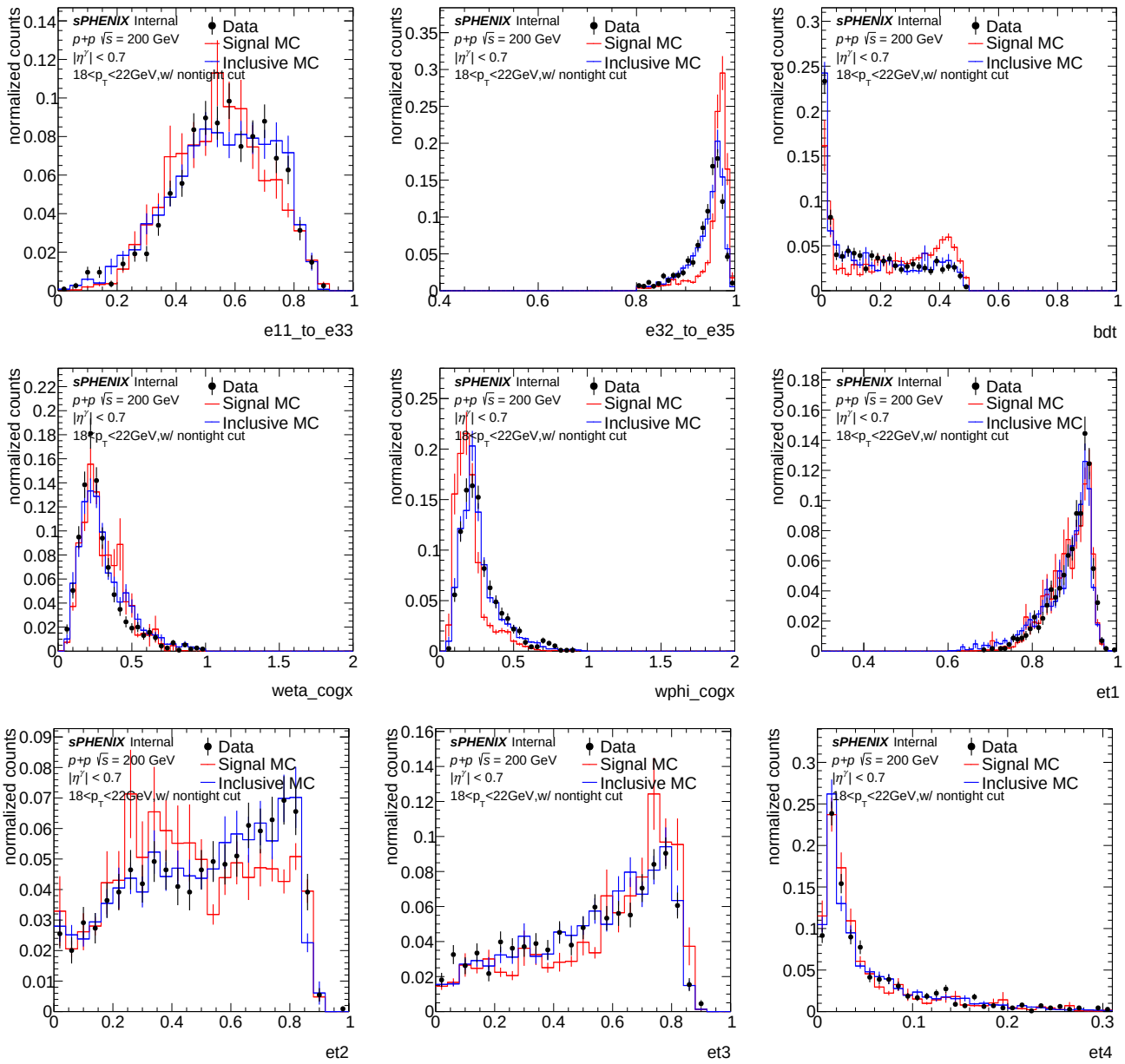

Figure 11: Shower shape distributions (Nominal, Data vs MC), $18 < E_T < 22$ GeV, non-tight (sideband).

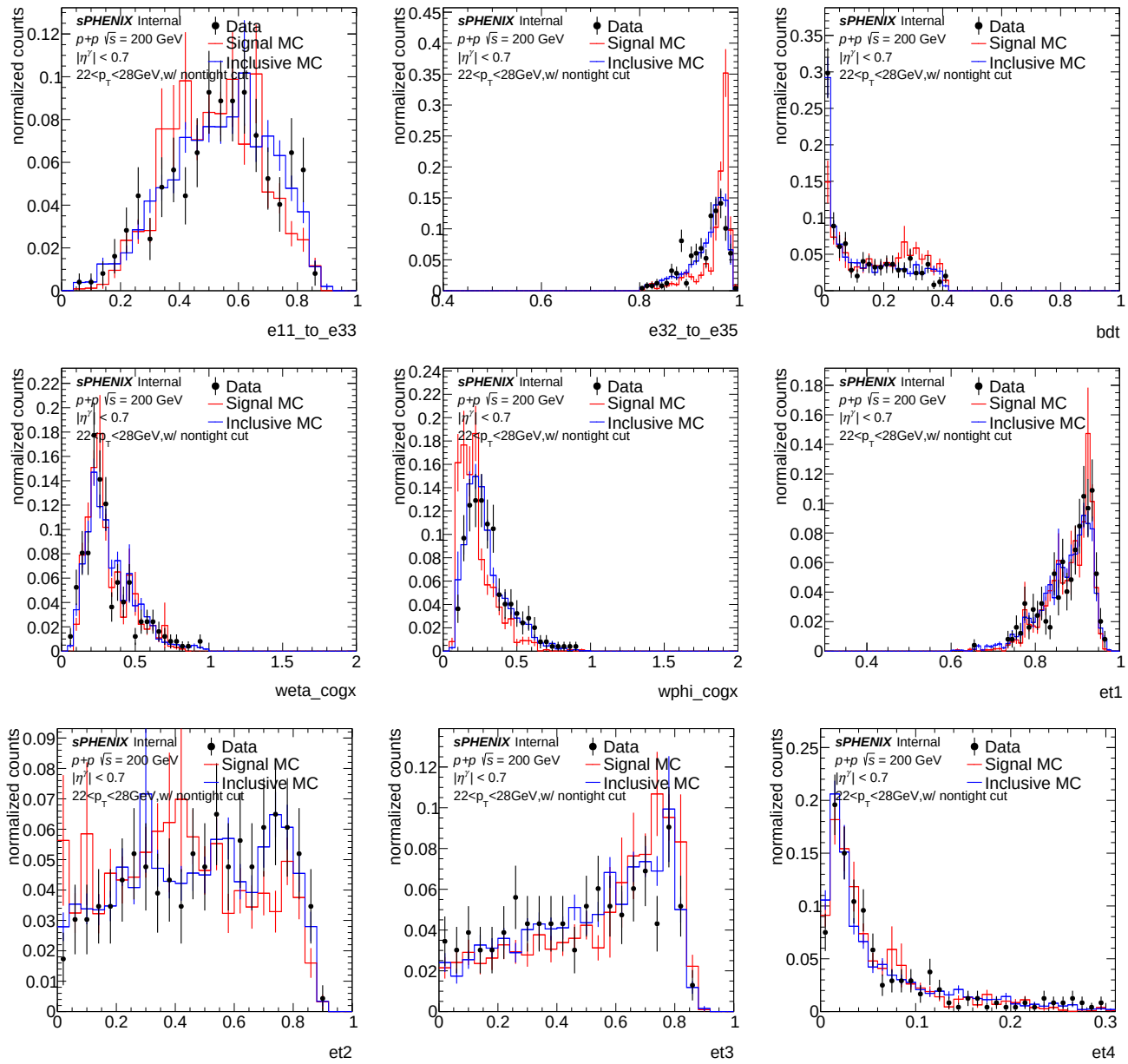

Figure 12: Shower shape distributions (Nominal, Data vs MC), $22 < E_T < 28$ GeV, non-tight (sideband).

1.2 Double-interaction mixed (0.813 × single + 0.187 × double)

1.2.1 no nbkg cut

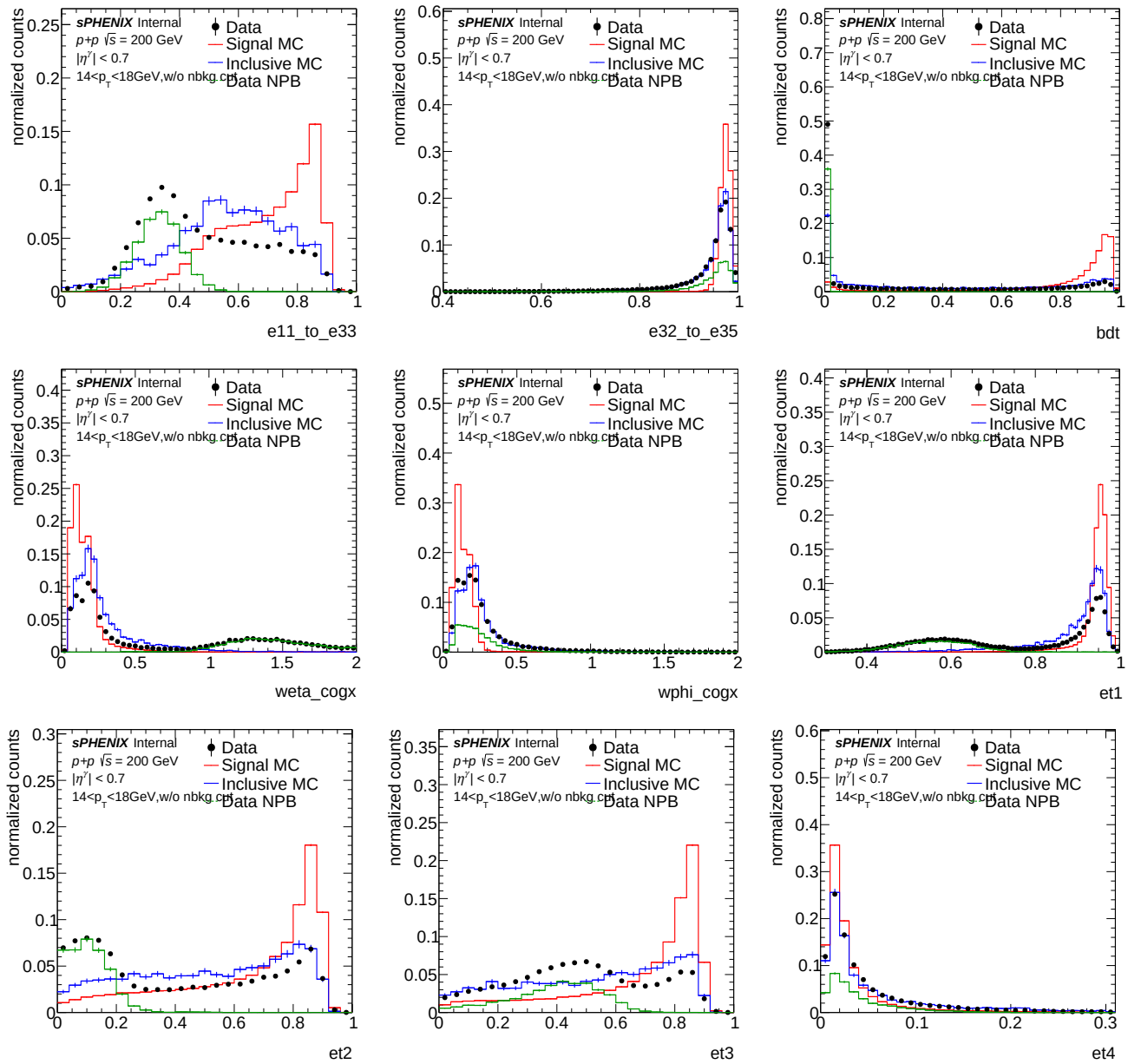

Figure 13: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $14 < E_T < 18$ GeV, no nbkg cut.

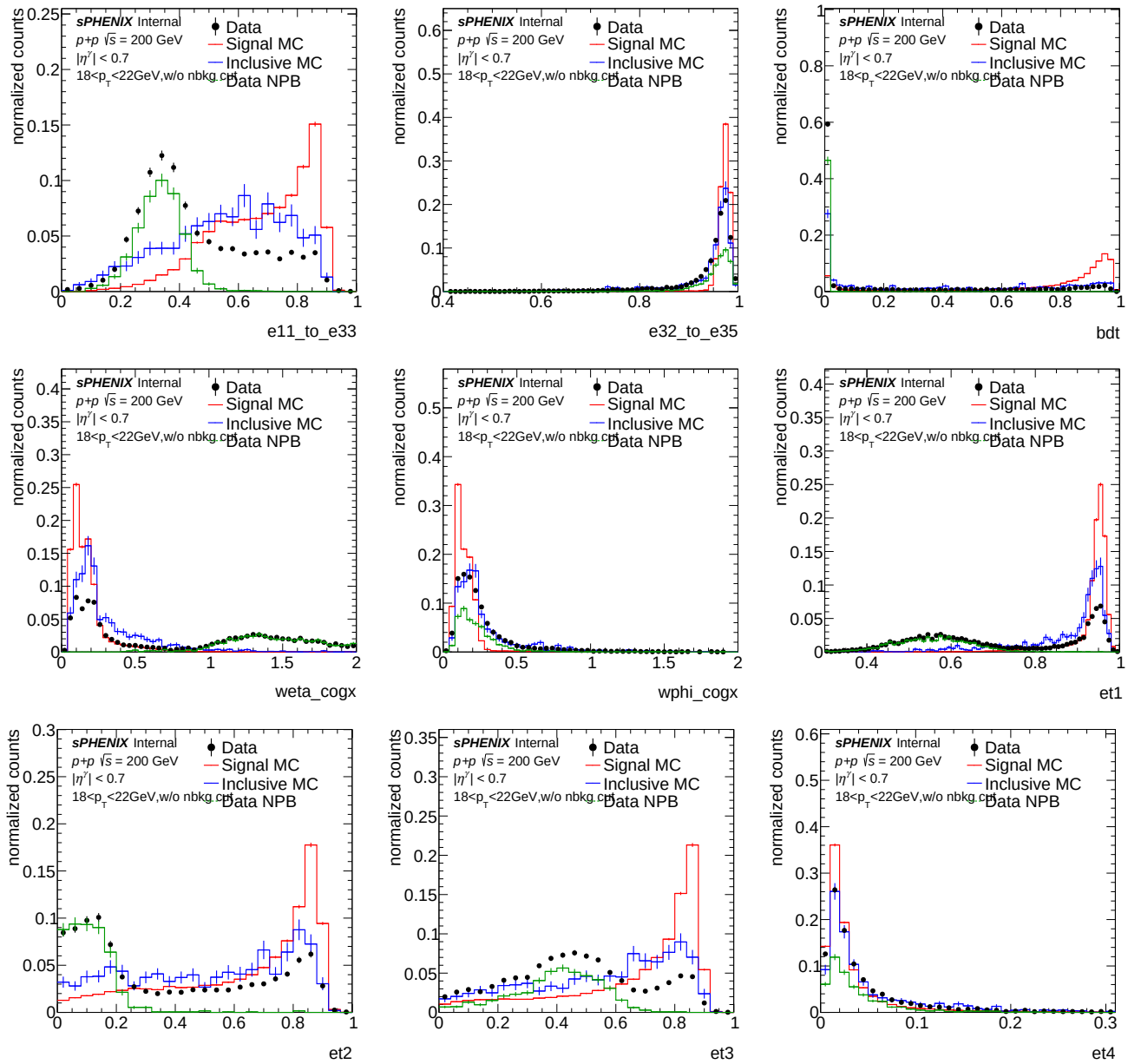

Figure 14: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $18 < E_T < 22$ GeV, no nbkg cut.

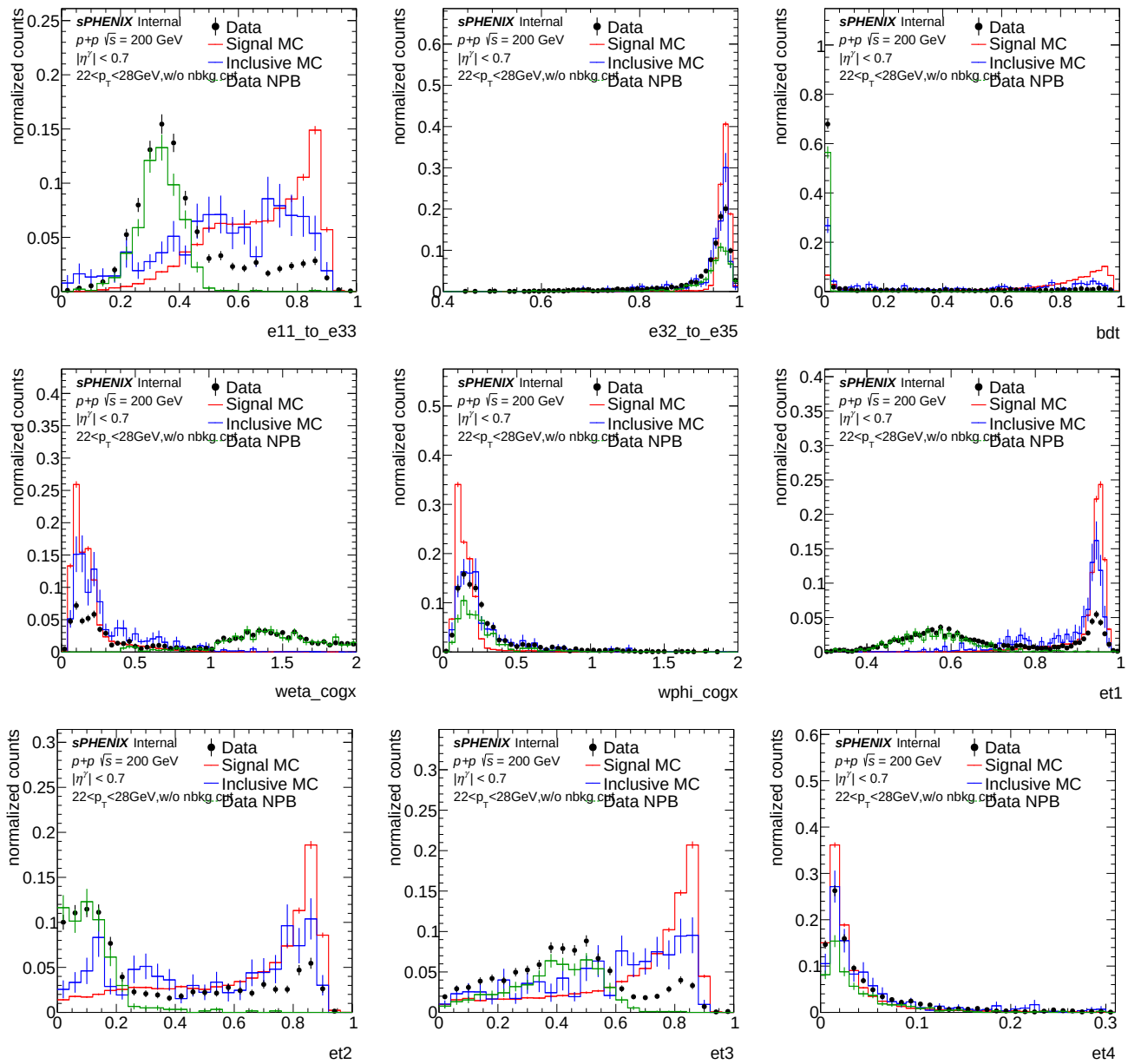

Figure 15: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $22 < E_T < 28$ GeV, no nbkg cut.

1.2.2 nbkg cut

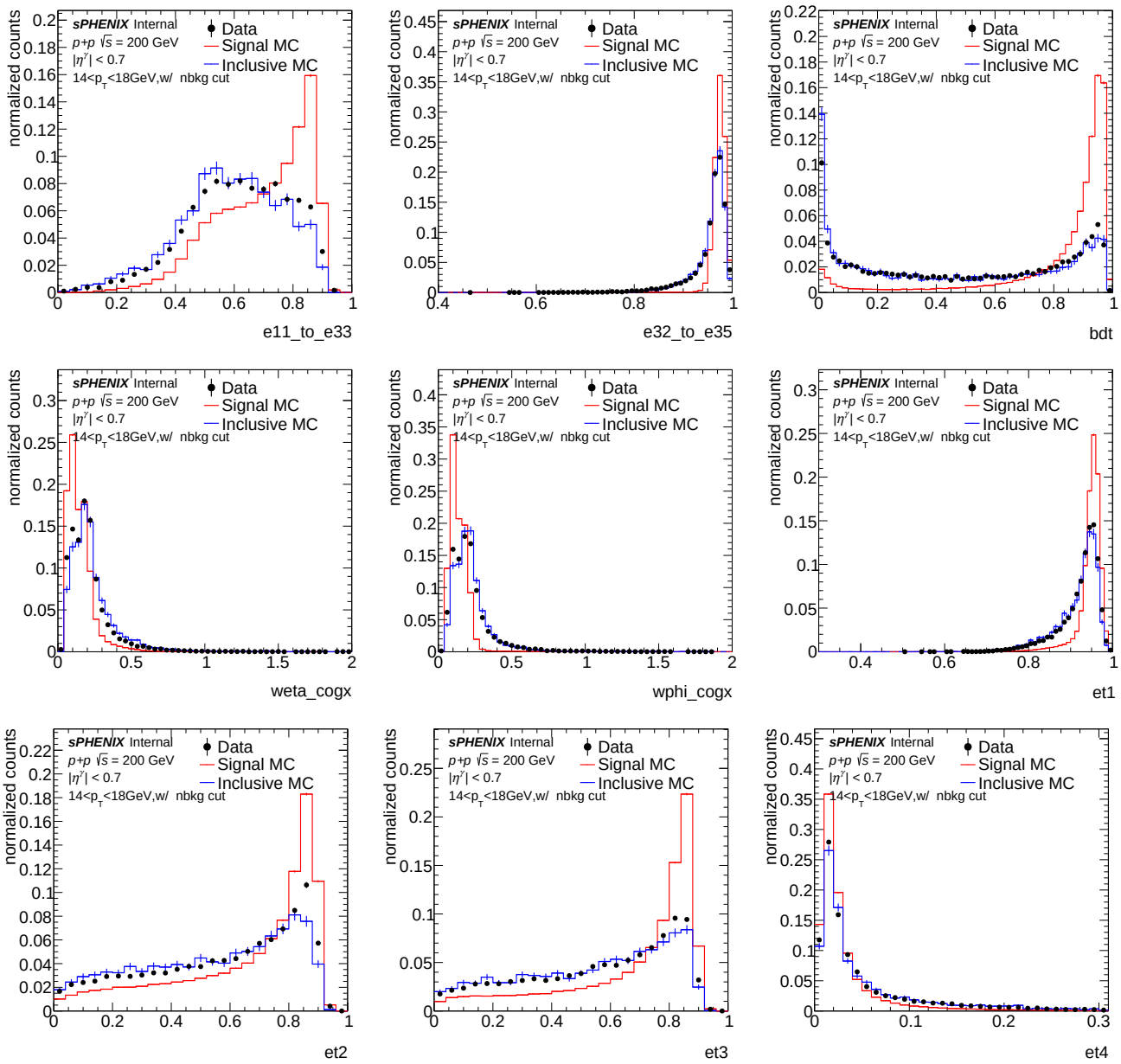

Figure 16: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $14 < E_T < 18$ GeV, nbkg cut.

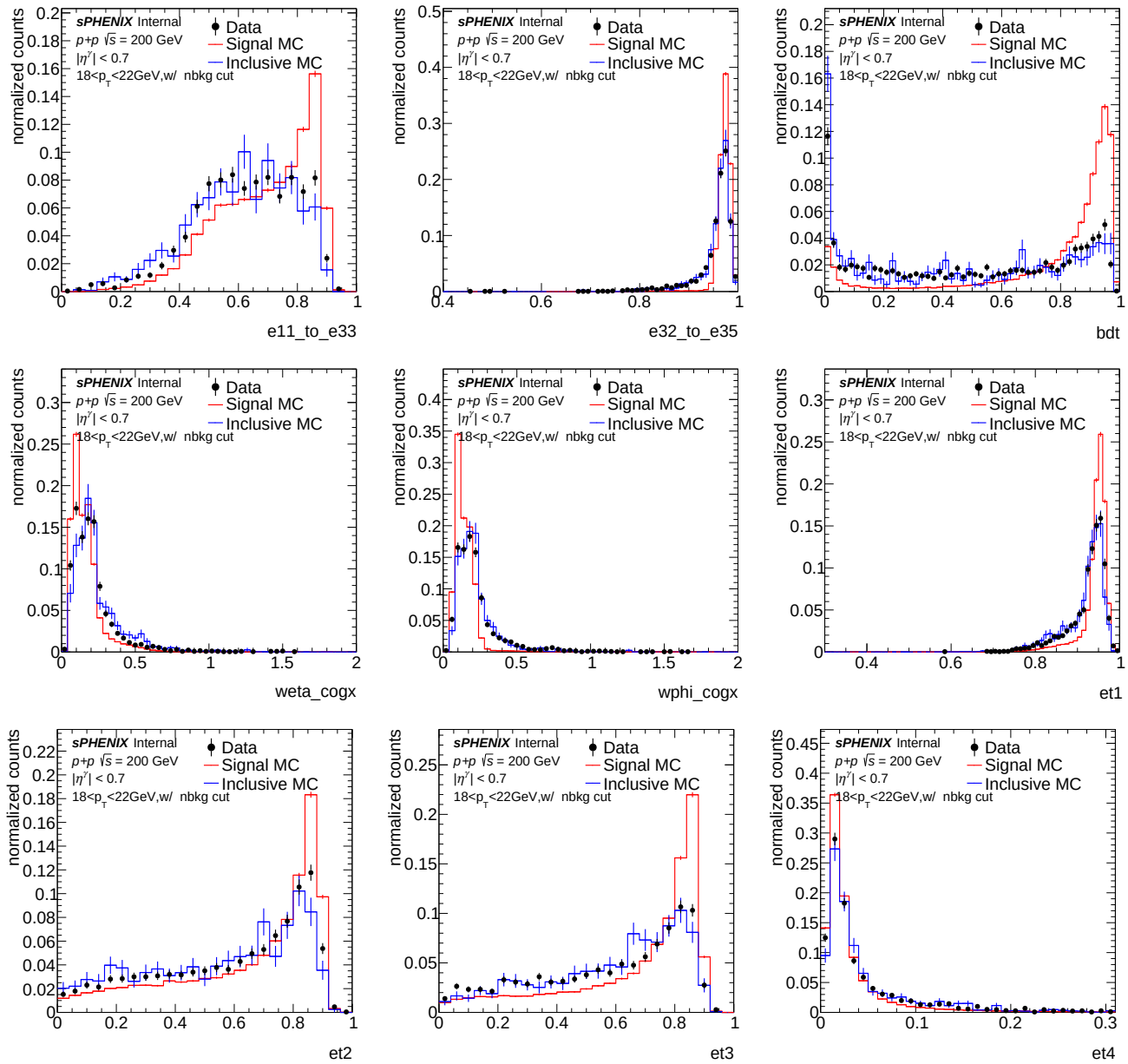

Figure 17: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $18 < E_T < 22$ GeV, nbkg cut.

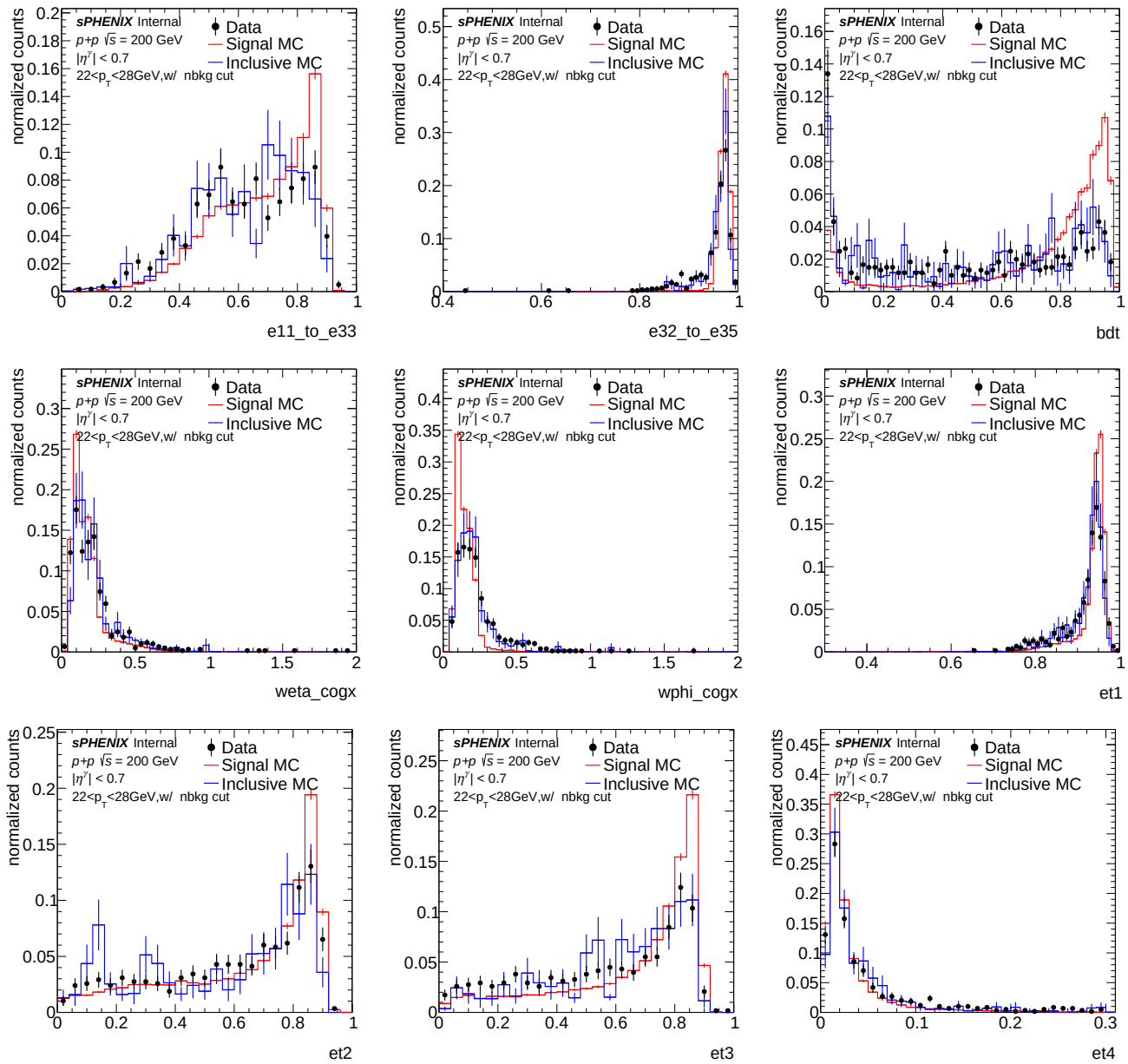

Figure 18: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $22 < E_T < 28$ GeV, nbkg cut.

1.2.3 tight photon ID

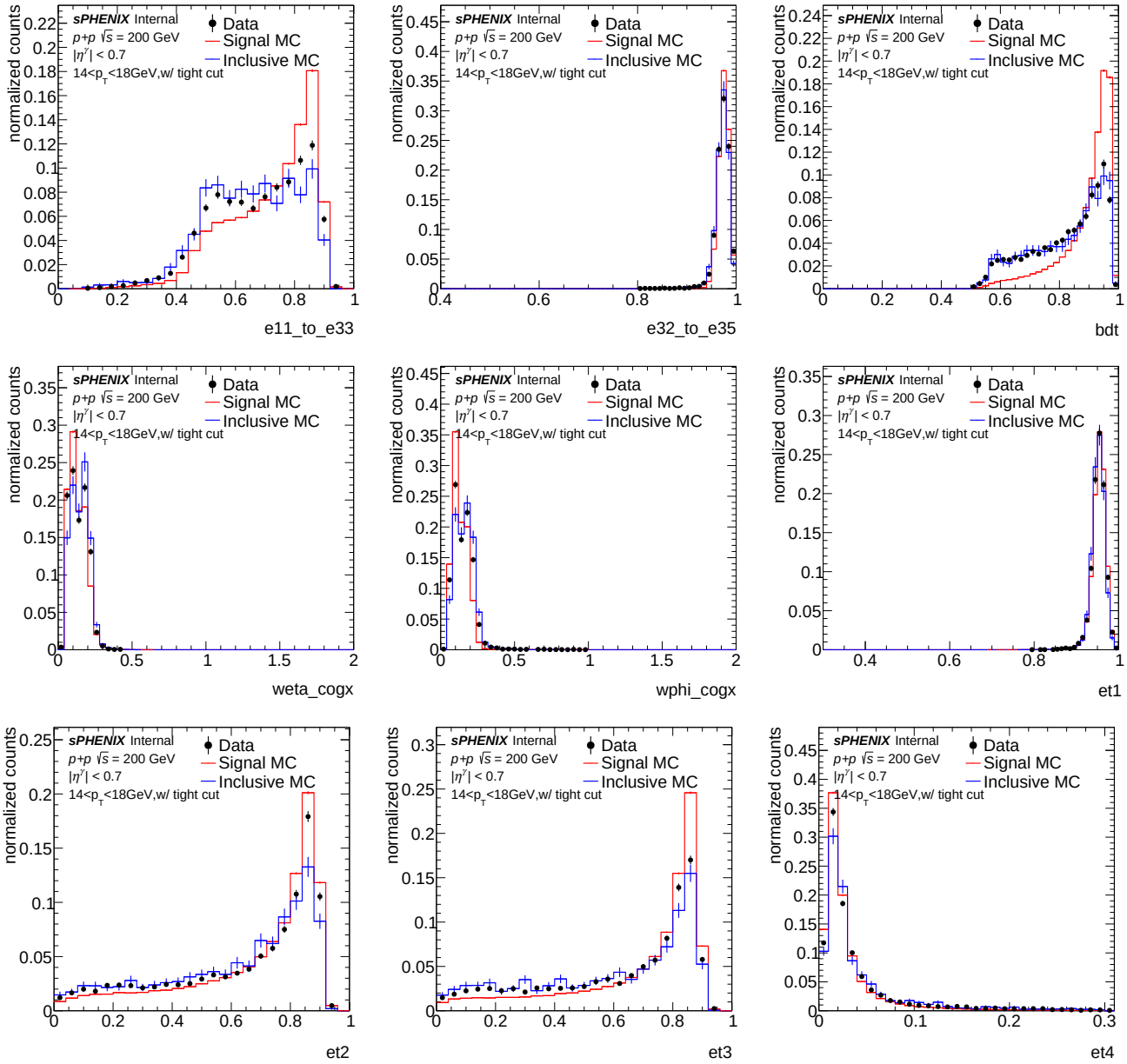

Figure 19: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $14 < E_T < 18$ GeV, tight photon ID.

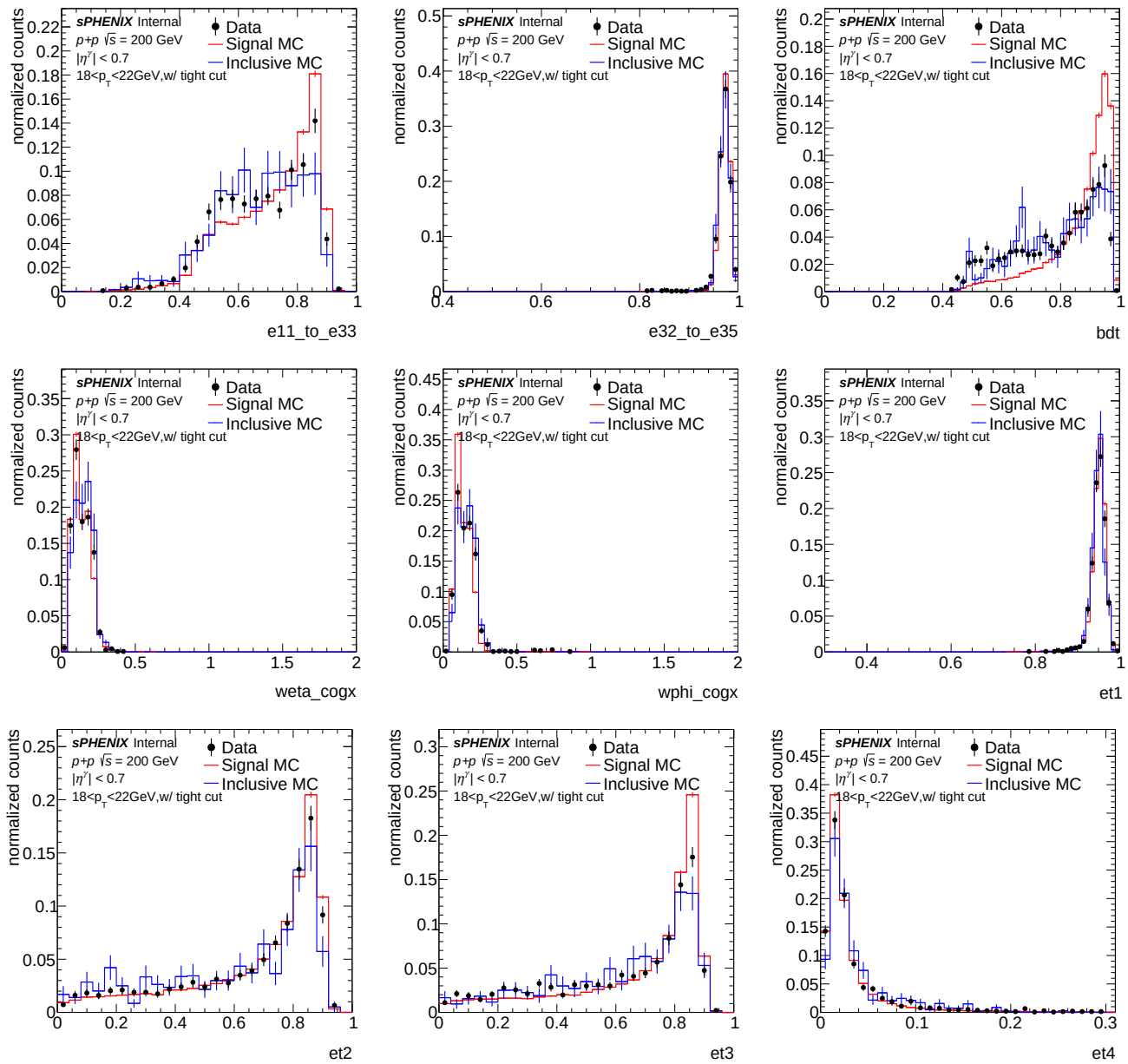

Figure 20: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $18 < E_T < 22$ GeV, tight photon ID.

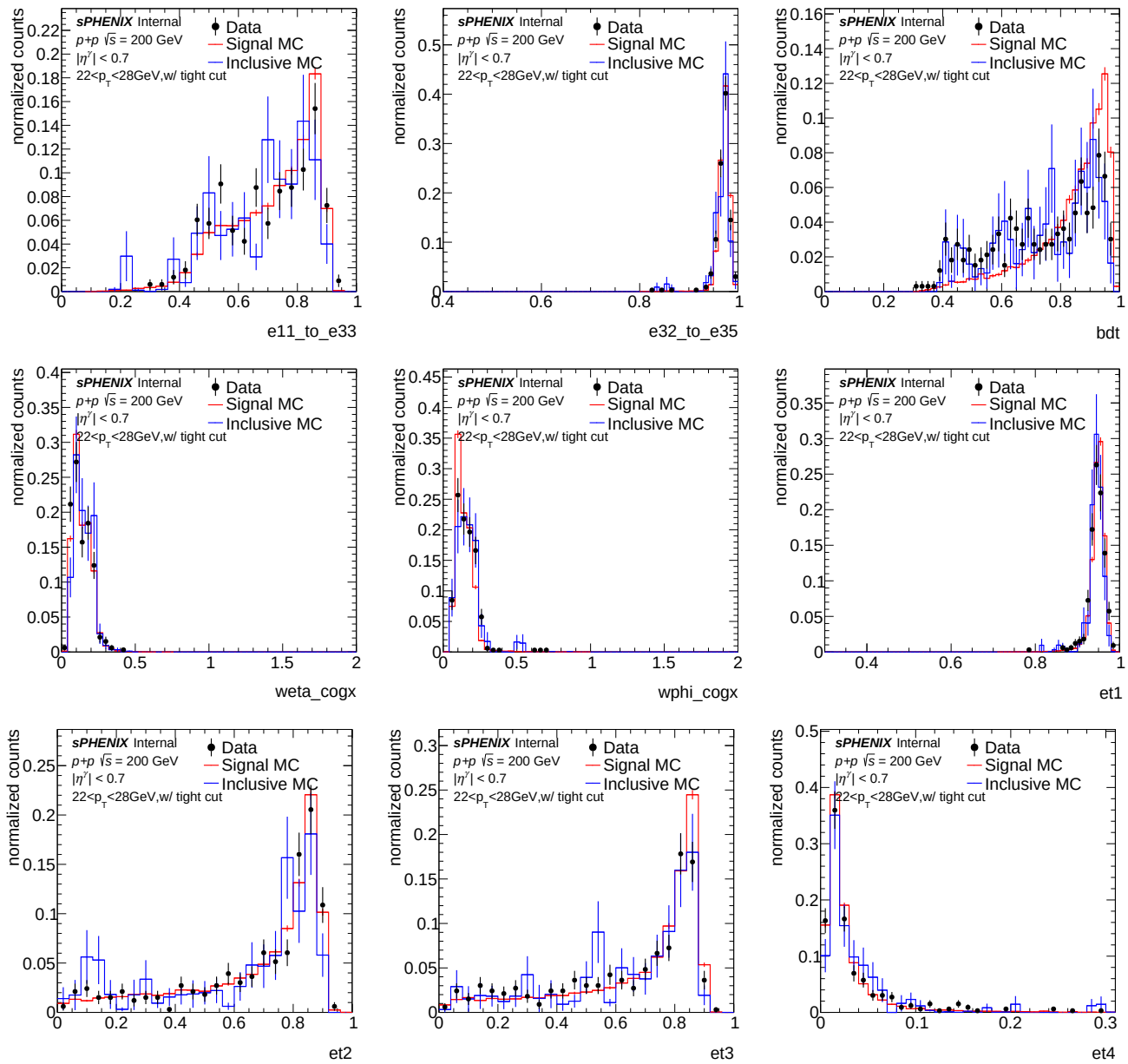

Figure 21: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $22 < E_T < 28$ GeV, tight photon ID.

1.2.4 non-tight (sideband)

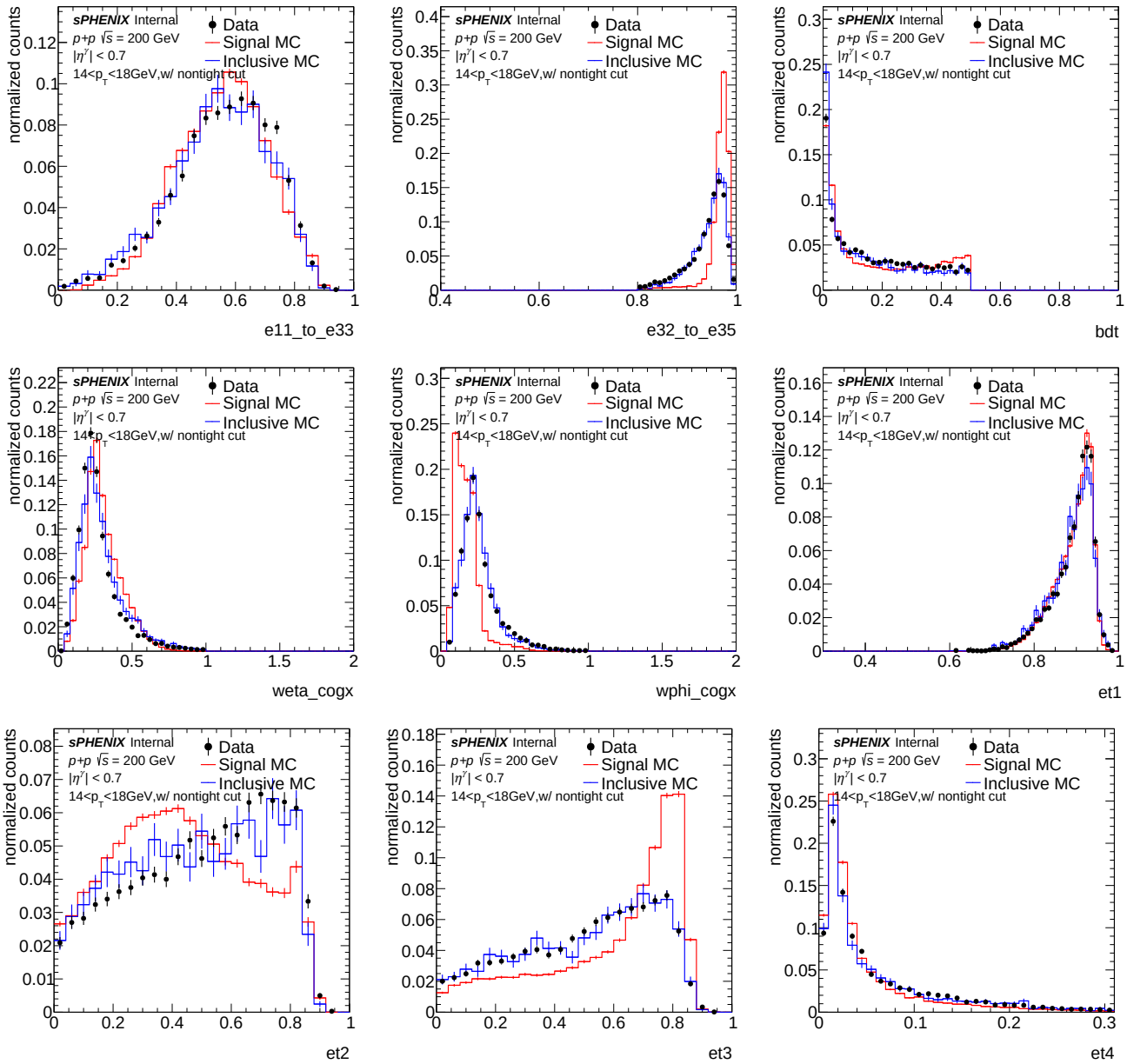

Figure 22: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $14 < E_T < 18$ GeV, non-tight (sideband).

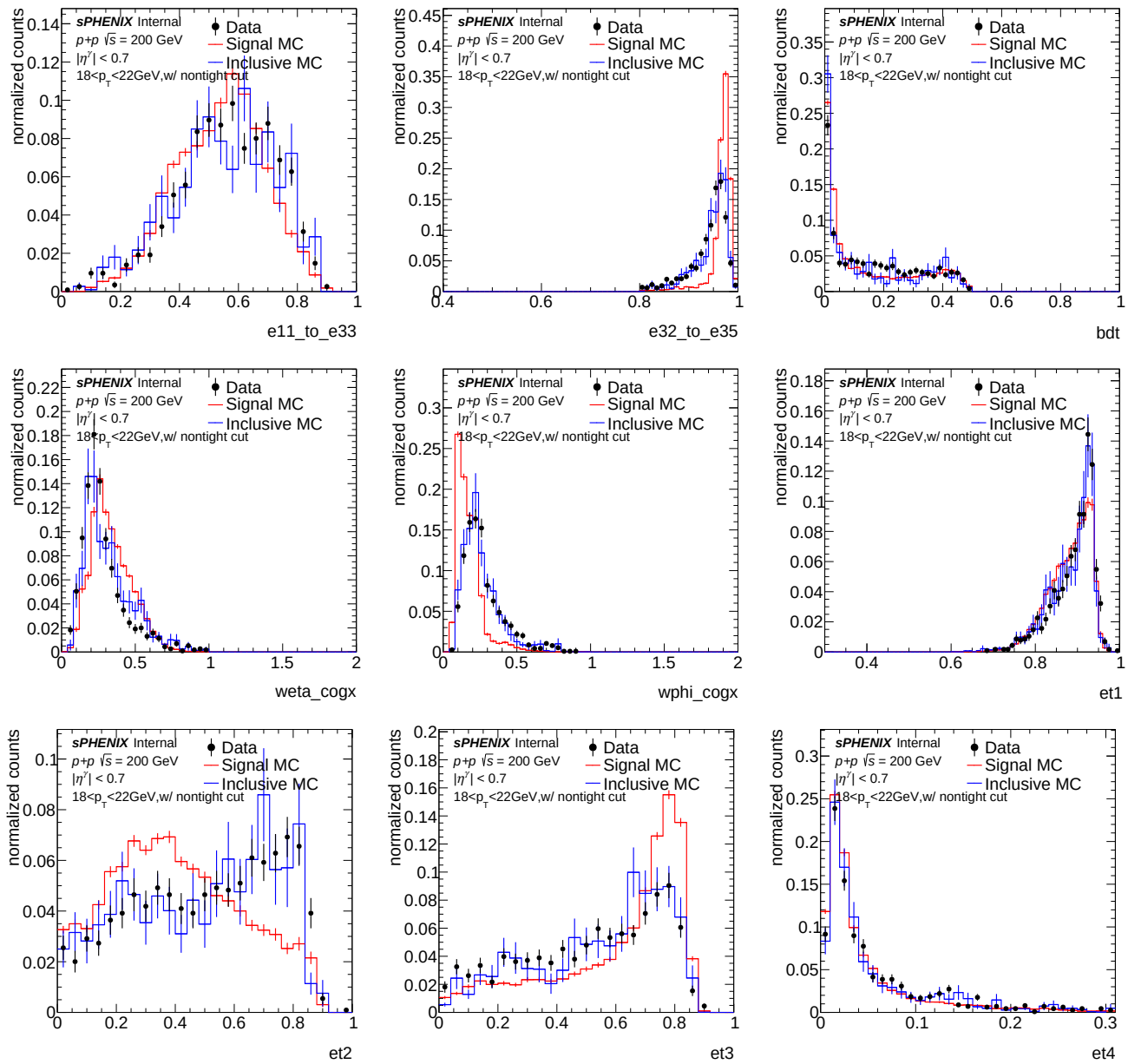

Figure 23: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $18 < E_T < 22$ GeV, non-tight (sideband).

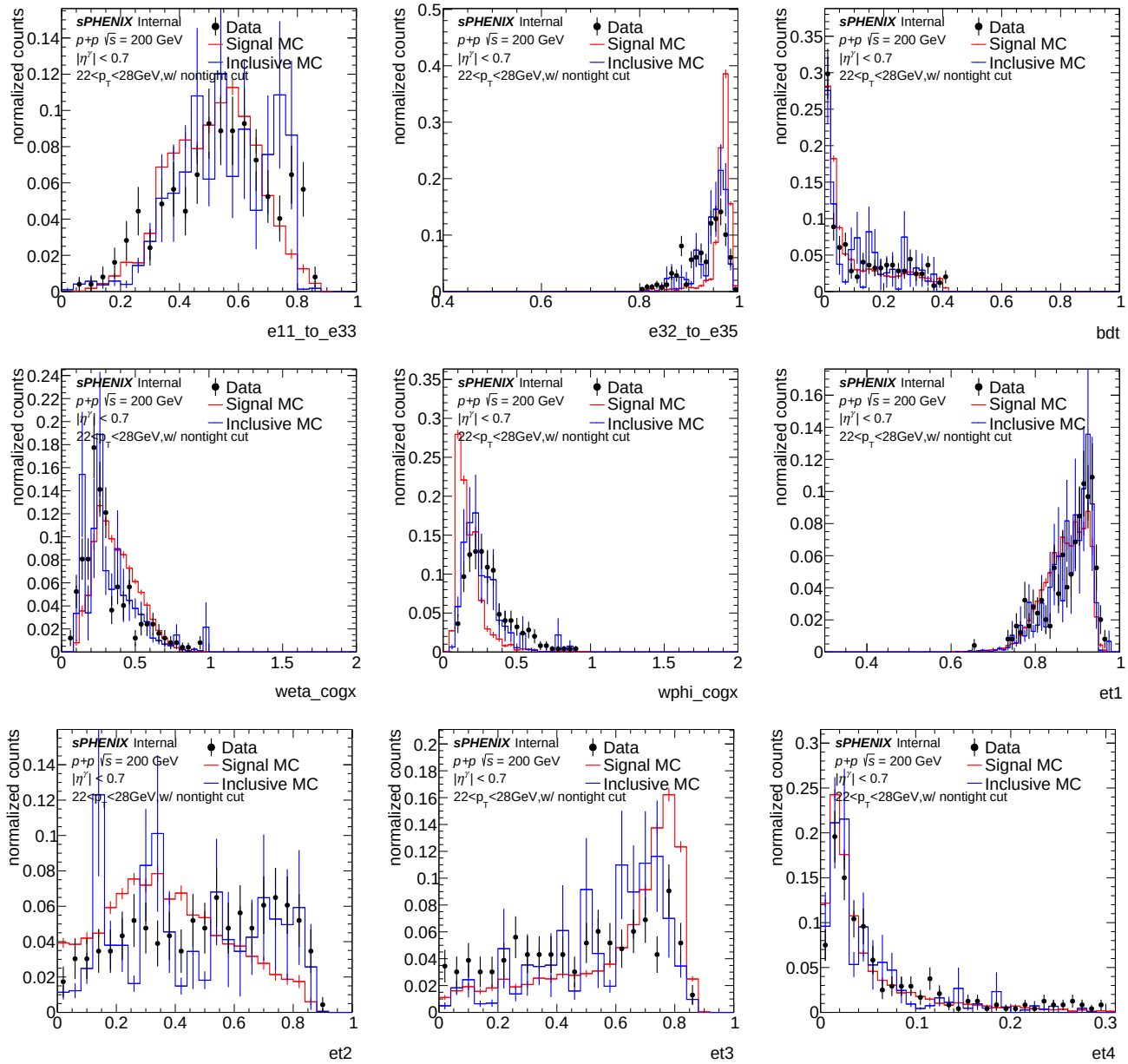

Figure 24: Shower shape distributions (Double-interaction mixed (0.813 × single + 0.187 × double), Data vs MC), $22 < E_T < 28$ GeV, non-tight (sideband).